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Economic Links from Bonds and Cross-Stock Return Predictability

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1 Big picture

- **Question.** Can economic links inferred from bond markets predict stock returns, and does this reveal segmentation between the corporate bond and equity markets?
- **Core idea.** Firms whose corporate bond ratings repeatedly move in the same direction around the same time are likely linked through common fundamentals, default-risk exposure, creditors, customers, or other bond-market relationships.
- **Main link measure.** The paper defines *rating-comovement links*: two firms are linked if their corporate bond ratings are updated in the same direction within a ± 10 day window at least three times over the past three years.
- **Main return signal.** For each focal firm, RCRet is the average monthly stock return of its rating-comovement peer firms. If peer firms react first to common information, RCRet should predict the focal firm's next-month stock return.
- **Main return result.** A monthly long-short stock strategy that buys high-RCRet focal firms and sells low-RCRet focal firms earns a value-weighted six-factor alpha around 0.45% per month and an equal-weighted alpha around 0.56% per month.
- **Why this is novel.** The return predictability is not subsumed by known equity-market links such as industry, geographic, customer, segment, technology, foreign-information, or shared-analyst momentum.
- **Fundamental link evidence.** Linked firms comove in profitability, sales, EPS, new capital, and default-probability measures such as O-score, Z-score, and expected default frequency.
- **Mechanism.** Market segmentation between bond and equity investors causes equity-market prices to underreact to bond-market information embedded in rating-comovement links.
- **Mechanism tests.** The effect is muted in the bond market, weakened when firms have dual holders that own both stocks and bonds, and visible in delayed equity analyst forecast revisions.
- **Economic takeaway.** Bond-market information can map out economically connected firms, and equity markets do not immediately integrate that information. Cross-asset segmentation therefore creates cross-stock return predictability.

2 Data and measurement

2.1 Bond-rating data and rating updates

- The main sample covers January 2001 through December 2019, matching the availability of TRACE corporate bond data.
- Ratings are transformed into numeric scores, with higher numbers corresponding to worse credit quality: AAA = 1, AA+ = 2, and so on through D.
- The paper focuses on credit-rating updates because rating agencies process both qualitative and quantitative information about firm fundamentals and credit risk.
- Fig. 1 shows that rating updates do not cluster strongly on particular days of the month or months of the year, making accidental calendar clustering less plausible.

Interpretation: Credit-rating changes are salient bond-market events. The paper uses repeated co-movement in these events to infer persistent economic links among firms.

2.2 Rating-comovement links

For two firms i and j , define a same-direction rating co-update if their ratings both move up or both move down within a ± 10 day calendar window. A rating-comovement link exists in month t if:

$$\#\{\text{same-direction rating co-updates between } i \text{ and } j \text{ in the past three years}\} \geq 3. \quad (1)$$

For each focal firm i in month t , let $\mathcal{P}_{i,t}$ denote its linked peer firms.

Interpretation: Requiring at least three co-updates in three years makes the link persistent rather than incidental. The link is constructed from public historical information, so the return strategy is tradeable.

2.3 Peer-return signal

The main cross-stock signal is:

$$\text{RCRet}_{i,t} = \frac{1}{|\mathcal{P}_{i,t}|} \sum_{j \in \mathcal{P}_{i,t}} R_{j,t}, \quad (2)$$

where $R_{j,t}$ is peer firm j 's stock return in month t .

Interpretation: RCRet summarizes value-relevant information arriving to linked peer firms. If equity investors underreact to the connection, peer returns in month t predict focal firm returns in month $t + 1$.

2.4 Stock and firm data

- The stock sample contains common stocks on NYSE, AMEX, and Nasdaq.
- The paper excludes stocks with share prices below \$1 and market capitalizations below the NYSE 10th percentile.
- Control variables include size, book-to-market, short-term reversal, momentum, asset growth, and gross profitability.
- Analyst forecasts come from I/B/E/S and are used to test delayed information incorporation by equity analysts.

Interpretation: The exclusions reduce microcap/liquidity contamination, while the controls address standard expected-return characteristics.

2.5 Bond returns

For a bond i , the TRACE-based monthly return is:

$$R_{i,t} = \frac{P_{i,t} + AI_{i,t} + C_{i,t}}{P_{i,t-1} + AI_{i,t-1}} - 1, \quad (3)$$

where P is clean price, AI is accrued interest, and C is coupon payment during the month.

Interpretation: Bond-return tests ask whether the information is already incorporated in the bond market. If the mechanism is bond-equity segmentation, cross-return predictability should be weak in bonds.

2.6 Dual holders

An asset-management company is a dual holder of a firm if it owns more than 1% of the firm's stock and more than 1% of the firm's bonds. A firm is classified as having dual holders if at least one asset manager meets both thresholds.

Interpretation: Dual holders can bridge information between equity and bond markets. If segmentation causes the anomaly, return predictability should be weaker when dual holders are present.

3 Models and empirical specifications

3.1 Cross-stock portfolio test

At the end of each month, focal firms are sorted into quintile portfolios based on $RCRet_{i,t}$. The paper computes next-month value-weighted and equal-weighted returns and estimates alphas under:

- Fama–French three-factor and four-factor models;
- Fama–French five-factor model;
- five-factor model plus momentum.

Interpretation: A positive high-minus-low alpha means that peer stock returns based on rating-comovement links predict focal stock returns after standard risk adjustment.

3.2 Fama–MacBeth regressions

The cross-sectional regression is:

$$R_{i,t+1} = a_t + b_t RCRet_{i,t} + \gamma'_t X_{i,t} + \varepsilon_{i,t+1}, \quad (4)$$

where $X_{i,t}$ contains size, book-to-market, reversal, momentum, asset growth, and gross profitability.

The paper also uses the ranking variable $RCRetRank_{i,t}$:

$$R_{i,t+1} = a_t + b_t RCRetRank_{i,t} + \gamma'_t X_{i,t} + \varepsilon_{i,t+1}. \quad (5)$$

Interpretation: These regressions test whether the signal predicts individual stock returns after controlling for firm characteristics.

3.3 Fundamental comovement regressions

At the focal-firm-by-year level, the paper estimates:

$$\Delta Fund_{i,y} = \alpha_i + \delta_y + \beta \overline{\Delta Fund}_{P_{i,y}} + u_{i,y}, \quad (6)$$

where $Fund$ is profit, sales, EPS, capital, O-score, Z-score, or expected default frequency.

Interpretation: A positive β means linked firms share common fundamentals, validating the economic-link interpretation of rating comovement.

3.4 Spanning regressions

The rating-comovement momentum factor RCMOM is regressed on other cross-firm momentum strategies:

$$RCMOM_t = \alpha + \beta' OtherMomentum_t + \theta' Factors_t + \varepsilon_t. \quad (7)$$

Interpretation: The intercept is the average abnormal return unexplained by other cross-firm momentum factors. A significant intercept means rating-comovement links add new information.

3.5 Analyst forecast revision regression

The analyst test is:

$$RevisionMean_{i,t+1} = \alpha + \beta RCRet_{i,t} + \gamma' X_{i,t} + \varepsilon_{i,t+1}. \quad (8)$$

Interpretation: If equity analysts learn slowly from bond-market links, peer returns should predict next-month consensus forecast revisions.

4 Identification and empirical design

4.1 What identifies the main effect

- Rating-comovement links are built from prior public rating events, not contemporaneous stock returns.
- The return signal uses peer stock returns in month t to predict focal stock returns in month $t + 1$.
- Portfolio alphas and Fama–MacBeth regressions control for standard equity expected-return effects.
- Spanning regressions compare the rating-comovement strategy to other known economic-link strategies.

Interpretation: The design identifies delayed diffusion of bond-link information into stock prices rather than ordinary stock return momentum alone.

4.2 Why fundamentals matter

- The paper first shows that rating-comovement peers comove in profitability, sales, EPS, capital, and credit-risk measures.
- This supports the interpretation that the links reflect real economic relationships, not just random rating-calendar overlap.
- Default-risk comovement is especially relevant because bond ratings primarily summarize credit risk.

Interpretation: Fundamentals are the bridge from rating co-updates to return predictability. Without real fundamental links, the trading strategy would be harder to interpret.

4.3 Why mechanism tests matter

- If the signal is truly bond-market information, it should be incorporated faster by bond prices than by stock prices.
- If segmentation is the friction, dual holders should reduce the delay.
- If equity analysts are slow to integrate bond-market links, RCRet should predict analyst forecast revisions.

Interpretation: The mechanism evidence distinguishes market segmentation from generic cross-firm momentum.

4.4 What remains imperfect

- Rating comovement may capture several underlying economic relationships rather than one single channel.
- Rating agencies may update firms strategically or in batches, although Fig. 1 and repeated-link requirements reduce this concern.
- Bond data availability restricts the sample mainly to larger, rated firms.
- The paper cannot directly observe investor attention or the internal information flow of dual holders.

Interpretation: The evidence is strongest as a joint case: links are fundamental, return predictability is robust, and the effect weakens where bond-equity segmentation should be lower.

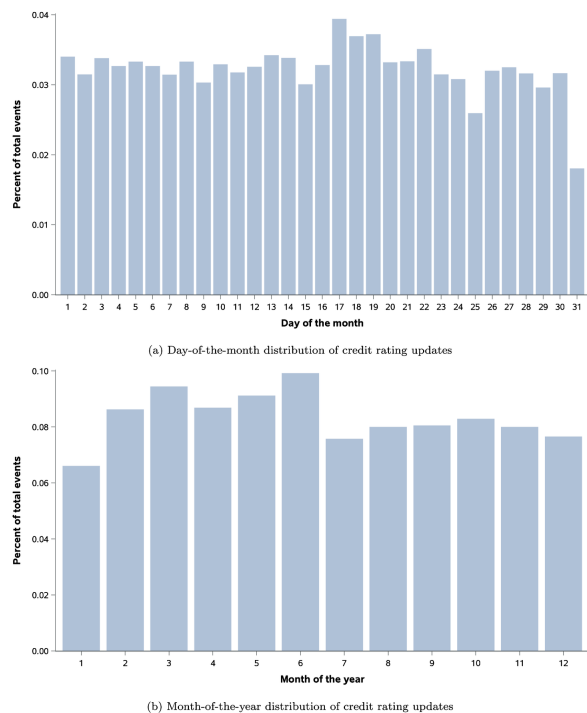
5 Tables and figures

5.1 Figure 1 and Table 1: rating update timing and sample summary

Read:

- Figure 1 shows that rating updates are dispersed across days of the month and months of the year, rather than heavily clustered at a fixed calendar date.
- Table 1 shows that focal and peer firms are large, value-tilted firms with average NYSE size percentiles above 70.
- An average focal firm has 27.37 peer firms, but the median is only 7, indicating a relatively sparse link network.
- The average firm receives less than one rating update per year, making repeated comovement unlikely by chance.

Economic message: Rating-comovement links are plausible signals of persistent economic connections rather than simple rating-calendar clustering.



(a) Figure 1: Rating update timing.

Table 1
Summary statistics.

	Mean	SD	P25	P50	P75
<i>Firm characteristics:</i>					
Focal firm size (NYSE percentile)	0.71	0.24	0.55	0.78	0.90
Peer firm average size (NYSE percentile)	0.77	0.14	0.72	0.80	0.86
Focal firm BM	0.75	0.82	0.32	0.58	0.91
Peer firm average BM	0.88	0.72	0.58	0.71	0.96
Focal firm numeric credit rating	9.54	3.43	7.00	9.00	12.00
Peer firm average numeric credit rating	9.25	2.29	7.98	8.80	10.25
<i>Rating-comovement link:</i>					
# Peer firms	27.37	48.75	2.00	7.00	27.00
# Rating updates per year (firms with ≥ 1 rating update)	0.79	1.79	0.00	0.00	1.00
# Rating updates per year (focal and peer firms)	1.05	2.03	0.00	0.00	2.00

This table reports summary statistics. Statistics on firm characteristics are generated based on focal-firm-by-month level observations of the sample period of Jan 2001 through Dec 2019. Firm-month observations with stock prices smaller than \$1 or market capitalization smaller than the NYSE 10th percentile value are excluded. For a focal firm in a given month, we compute the average value of the characteristics of its peer firms (peer-firm average characteristics), and we count the number of peer firms connected to the focal firm in the month as # Peer firms. Size is market capitalization. BM is the book-to-market ratio. Numeric credit-rating increases with credit risk (1 = AAA, 2 = AA+, ..., 10 = BBB-, ..., 22 = D). A firm's numeric credit rating in a month is computed as the average numeric credit rating of its outstanding corporate bond issues. The statistics on rating update frequency are produced based on firm-year level observations in the sample period of 1998 through 2019. For each firm in each year, # Rating updates per year is the number of times the firms' bond issues were updated in this year (0 indicates no rating update). We report the frequency of rating updates based on the sample of firms with at least one rating update separately from those based on the sample of focal and peer firms in the rating-comovement linkage.

(b) Table 1: Summary statistics.

5.2 Table 2 and Table 3: fundamentals and stock-return predictability

Read:

- Table 2 shows significant comovement in fundamentals between focal and peer firms.
- Peer profit growth predicts focal profit growth with a coefficient of 0.365; peer sales, EPS, and new capital also predict corresponding focal variables.
- Default-probability measures also comove: peer O-score, Z-score, and EDF changes predict focal changes.

- Table 3 shows strong stock-return predictability. The value-weighted high-minus-low portfolio has a 0.45% monthly FF5+UMD alpha.
- The equal-weighted high-minus-low portfolio has a 0.56% monthly FF5+UMD alpha.

Economic message: The links are economically meaningful and tradable: peer firms share fundamentals, and peer stock returns forecast focal stock returns.

Table 2
Fundamentals comovement.

Panel A: Growth Comovement				
DepVar:	(1) Δ Profit	(2) Δ Sales	(3) Δ EPS	(4) New Capital
Peer Δ Profit	0.365*** (5.77)			
Peer Δ Sales		0.163** (2.72)		
Peer Δ EPS			0.089*** (8.16)	
Peer New Capital				0.110*** (3.82)
Firm FE	Y	Y	Y	Y
Time FE	Y	Y	Y	Y
No. Obs.	5,620	5,681	5,678	5,154
Adj. R ²	0.113	0.191	0.103	0.117
Panel B: Default Probability Comovement				
DepVar:	(1) Δ O-score	(2) Δ Z-score	(3) Δ EDF	
Peer Δ O-score	0.098** (2.39)			
Peer Δ Z-score		0.114** (2.36)		
Peer Δ EDF			0.226*** (5.52)	
Firm FE	Y	Y	Y	
Time FE	Y	Y	Y	
No. Obs.	4,321	4,290	5,672	
Adj. R ²	0.025	0.171	0.172	

This table reports results from panel regressions of focal firms' fundamental variables on the average value of its peer firms' contemporaneous fundamental variables. The unit of observation is at the firm-by-year level, and the sample period is from 2001 to 2019. Panel A reports results based on fundamental variables concerning firm growth. In column (1), profitability is measured by the operating income before depreciation (Compustat item: OIBDP). Δ Profit in a given year is defined as the change in profitability from the previous year to the current year, scaled by total assets. In column (2), Δ Sales is the year-to-year change in sales scaled by total assets. In column (3), Δ EPS is the year-to-year change in EPS standardized by the lagged stock price. In column (4), new capital is defined as the sum of net equity issuance plus net debt issuance scaled by lagged total assets. Peer Δ Profit is the average profit growth among peer firms in the same year. Peer Δ Sales is the average sales growth among peer firms in the same year. Peer Δ EPS is the average EPS growth among peer firms in the same year.

(a) Table 2: Fundamentals comovement.

Table 3
Performance of stock portfolios sorted by $RCRet$.

Panel A: Value Weighted				
RCRet Portfolios	Excess Ret	FFC4	FF5	FF5+UMD
1 (Low)	0.61 (1.55)	-0.07 (-0.59)	-0.15 (-1.11)	-0.15 (-1.10)
2	0.56 (1.38)	-0.09 (-0.86)	-0.20 (-1.60)	-0.20 (-1.60)
3	0.71 (1.89)	0.11 (1.08)	0.03 (0.27)	0.03 (0.25)
4	0.78 (2.19)	0.20 (1.72)	0.07 (0.65)	0.06 (0.63)
5 (High)	1.06 (3.33)	0.45 (6.46)	0.31 (3.54)	0.30 (2.96)
5-1	0.45 (3.20)	0.52 (3.95)	0.45 (3.39)	0.45 (3.47)
Panel B: Equal Weighted				
RCRet Portfolios	Excess Ret	FFC4	FF5	FF5+UMD
1 (Low)	0.71 (1.60)	-0.08 (-0.65)	-0.21 (-1.72)	-0.21 (-1.57)
2	0.78 (1.83)	0.03 (0.28)	-0.12 (-1.07)	-0.12 (-1.06)
3	0.88 (2.21)	0.20 (1.58)	0.05 (0.37)	0.04 (0.37)
4	0.99 (2.48)	0.33 (2.59)	0.19 (1.42)	0.18 (1.74)
5 (High)	1.21 (3.03)	0.49 (5.44)	0.35 (3.68)	0.35 (3.51)
5-1	0.50 (3.41)	0.57 (4.34)	0.56 (4.20)	0.56 (4.65)

This table reports the performance of stock portfolios sorted by $RCRet$ in the holding period from Jan 2001 through Dec 2019. At each month-end, stocks are sorted into quintiles by their $RCRet$ in that month. The quintile portfolios are held over the next month. Stocks with share prices below \$1 or market capitalizations below the NYSE 10th percentile value are excluded at the portfolio formation. We report average monthly portfolio excess returns and portfolio alphas under the Fama-French-Carhart four-factor model (FFC4), the Fama-French five-factor model (FF5), and the Fama-French five-factor model augmented with momentum factor (FF5+UMD). Panel A reports the performance of value-weighted portfolios, and Panel B reports the performance of equal-weighted portfolios. t -statistics in parentheses are computed based on standard errors with Newey-West correction of 12 lags.

(b) Table 3: Stock portfolios sorted by $RCRet$.

5.3 Table 4 and Table 5: regressions and information environment

Read:

- Table 4 shows that $RCRet$ predicts one-month-ahead focal firm returns in Fama-MacBeth regressions.
- The raw $RCRet$ coefficient is 0.0323 without controls and 0.0202 with controls.
- The quintile rank version is also significant, with a coefficient of 0.0012 without controls and 0.0007 with controls.
- Table 5 shows that predictability is stronger when peer firms are large and focal firms are small.
- In the large-peer/small-focal subsample, the FF5+UMD alpha is 0.69% per month; in the opposite direction, it is only 0.34%.

Economic message: Information appears to diffuse from better-information-environment firms to worse-information-environment firms, consistent with delayed information transmission.

Table 4
Fama-MacBeth regressions.

	(1)	(2)	(3)	(4)
RCRet	0.0323*** (3.29)	0.0202** (2.25)		
RCRet Rank			0.0012*** (3.00)	0.0007** (2.04)
Size		-0.0010* (-1.66)		-0.0010* (-1.68)
BM		-0.0001 (-0.06)		-0.0002 (-0.08)
RVSL		0.0050 (0.51)		0.0050 (0.51)
MOM		-0.0031 (-0.63)		-0.0033 (-0.66)
AG		-0.0078** (-2.36)		-0.0079** (-2.33)
GP		0.0059 (1.29)		0.0058 (1.24)
No. Obs.	65,453	65,453	65,453	65,453
Adj. R ²	0.004	0.108	0.003	0.107

This table reports results from Fama-MacBeth forecasting regressions of future stock returns. The sample period is from Jan 2001 through Dec 2019. The dependent variable is stock returns in month $t+1$, and the key independent variable is $RCRet$ or the quintile ranking of $RCRet$ ($RCRetRank$) in month t . Stocks with share prices below \$1 or market

(a) Table 4: Fama-MacBeth regressions.

Table 5
Information environment and return predictability.

Panel A: Large Peer Firms and Small Focal Firms				
RCRet Portfolios	Excess Ret	FFC4	FF5	FF5+UMD
1 (Low)	0.69 (1.37)	-0.16 (-0.91)	-0.29 (-1.70)	-0.30 (-1.60)
2	1.03 (2.46)	0.25 (1.39)	0.09 (0.44)	0.08 (0.41)
3	1.07 (2.50)	0.35 (1.83)	0.19 (1.07)	0.18 (0.98)
4	1.07 (1.98)	0.26 (1.17)	0.12 (0.44)	0.11 (0.46)
5 (High)	1.42 (2.80)	0.67 (2.98)	0.40 (1.74)	0.39 (2.62)
5-1	0.73 (2.11)	0.83 (2.76)	0.70 (2.17)	0.69 (2.71)
Panel B: Small Peer Firms and Large Focal Firms				
RCRet Portfolios	Excess Ret	FFC4	FF5	FF5+UMD
1 (Low)	0.47 (1.05)	-0.16 (-0.81)	-0.25 (-1.14)	-0.25 (-1.14)
2	0.47 (1.04)	-0.15 (-0.89)	-0.22 (-1.31)	-0.22 (-1.31)
3	0.66 (2.01)	0.11 (0.90)	0.05 (0.43)	0.05 (0.43)
4	0.58 (1.47)	0.02 (0.16)	-0.12 (-0.75)	-0.12 (-0.73)
5 (High)	0.80 (2.73)	0.26 (3.49)	0.09 (0.87)	0.09 (0.70)
5-1	0.33 (1.63)	0.41 (2.05)	0.35 (1.75)	0.34 (1.73)

This table reports results from a sub-sample analysis on the return predictability of $RCRet$ from Jan 2001 through Dec 2019. At each month end, we identify the median

(b) Table 5: Information environment.

5.4 Table 6: spanning regressions

Read:

- Table 6 regresses rating-comovement momentum on other cross-firm momentum strategies and standard factors.
- The rating-comovement strategy earns about 48 bps per month with FF5+UMD controls.
- After controlling for composite bond-market momentum, shared-analyst momentum, industry momentum, geographic momentum, customer momentum, segment momentum, technology momentum, and foreign-information momentum, the alpha remains positive at about 40 bps.

Economic message: Rating-comovement momentum is not simply another version of known equity-market economic-link strategies. It captures information specific to the bond market.

Table 6
Spanning regressions.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Intercept	0.482*** (3.74)	0.415*** (3.14)	0.472*** (3.60)	0.474*** (3.70)	0.469*** (3.70)	0.415*** (3.02)	0.461*** (3.47)	0.499*** (3.76)	0.461*** (3.49)	0.402*** (2.64)
Composite BondMKT MOM		0.198** (2.35)								0.030 (0.40)
Shared-Analyst MOM			0.112*** (3.09)							0.010 (0.22)
Industry MOM				0.150*** (3.74)						0.056 (1.63)
Geographic MOM					0.070 (1.45)					-0.009 (-0.22)
Customer MOM						0.108** (2.35)				0.068 (1.63)
Segment MOM							0.153*** (5.19)			0.111*** (3.34)
Technology MOM								0.058* (1.84)		-0.044 (-0.95)
Foreign Info MOM									0.075** (1.98)	-0.011 (-0.42)
FF5+UMD	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
No. Obs.	222	222	222	222	222	222	222	222	222	222
Adj. R ²	0.119	0.179	0.182	0.191	0.127	0.173	0.205	0.13	0.136	0.245

This table reports results from the spanning regressions of rating-comovement momentum long-short returns on other cross-momentum long-short returns, controlling for Fama and French (2015) five factors and the UMD factor. The construction of these factors is described in Internet Appendix B. The sample period is from July 2001 through December 2019 due to the availability of composite bond market cross-momentum factor returns. The dependent variable is the return of rating-comovement momentum factor. The key independent variables in columns (2)–(9) are the contemporaneous returns of composite bond market cross-momentum, shared analyst momentum (Ali and Hirschleifer, 2020), industry momentum (Moskowitz and Grinblatt, 1999), geographic momentum (Parsons et al., 2020), customer momentum (Cohen and Frazzini, 2008), segment momentum (Cohen and Lou, 2012), technology momentum (Lee et al., 2019), and foreign information momentum (Huang, 2015), respectively. Contemporaneous Fama and French (2015) five factors and momentum factor (UMD) are controlled for. *t*-statistics in parentheses are computed based on standard errors with Newey–West correction of 12 lags. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

5.5 Table 7 and Table 8: cross-bond predictability and dual holders

Read:

- Table 7 shows no significant cross-bond return predictability based on rating-comovement links.
- The high-minus-low bond portfolio alphas are small and statistically insignificant.
- Table 8 shows that stock predictability is strong when firms do not have dual holders, but much weaker when dual holders exist.
- Without dual holders, the high-minus-low FF5+UMD alpha is 0.48% per month; with dual holders, it falls to 0.29% and is weaker statistically.

Economic message: The bond market appears to incorporate bond-market information quickly, while stock markets are slower. Dual holders reduce segmentation and therefore reduce predictability.

Table 7
Performance of bond portfolios.

	(1) Equal Weighted	(2)	(3) Value Weighted	(4)
1 (Low)	0.06 (0.80)	0.04 (0.47)	-0.01 (-0.29)	-0.01 (-0.25)
2	-0.02 (-0.34)	-0.02 (-0.27)	-0.06 (-1.22)	-0.07 (-1.39)
3	0.01 (0.11)	-0.05 (-0.63)	-0.02 (-0.35)	-0.10 (-1.85)
4	0.08 (1.31)	0.04 (0.61)	0.00 (-0.02)	-0.06 (-1.05)
5 (High)	0.19 (2.58)	0.21 (2.76)	0.10 (1.24)	0.05 (0.68)
5–1	0.13 (1.23)	0.17 (1.41)	0.11 (1.47)	0.06 (1.41)

This table reports the performance of bond portfolios sorted by $RCRet_{Bond}$ in the holding period from Jan 2001 through Dec 2019. For each firm in each month, we compute firm-level bond returns as the average returns of outstanding corporate bonds issued by the firm; equal-weighted returns are reported in columns (1) and (3) and offering-amount-weighted returns are reported in columns (2) and (4). Then, for a focal firm in a given month, $RCRet_{Bond}$ is defined as the average firm-level bond returns across its peer firms in that month. At each month-end, focal firms are sorted into quintiles by their $RCRet_{Bond}$ in that month. We track the average firm-level bond returns for the quintile portfolios over the next month; equal-weighted returns are reported in columns (1) and (2) and value-weighted returns are reported in columns (3) and (4). We report the average monthly portfolio alpha relative to the corporate bond market factor, where the corporate bond market factor return is the value-weighted returns of all outstanding US corporate bonds in excess of the risk-free rate. *t*-statistics in parentheses are computed based on standard errors with Newey–West correction of 12 lags.

(a) Table 7: Bond portfolios.

Table 8
Dual holder and return predictability.

Dual Holder:	Excess Returns		FF5+UMD Alpha	
	No	Yes	No	Yes
Low RCRet	0.38 (1.04)	0.42 (0.97)	-0.34 (-2.95)	-0.28 (-3.03)
High RCRet	0.88 (2.88)	0.69 (1.93)	0.14 (1.47)	0.01 (0.05)
High–Low	0.50 (3.47)	0.28 (1.23)	0.48 (3.22)	0.29 (1.74)

This table reports the performance of stock portfolios two-way sorted by $RCRet$ and by the existence of dual holders during the holding period of Jan 2001 through Dec 2019. A firm has dual holders if at least one asset management company owns more than 1% of the firm's outstanding stock shares and more than 1% of the firm's outstanding bond issues. At each month-end, we split the focal firms into two groups based on the existence of dual holders, and we also split the focal firms into two even groups by their $RCRet$ this month. The two-way sorted portfolios are held over the next month, and value-weighted portfolio returns are computed. Average monthly excess returns and FF5+UMD alphas are reported. *t*-statistics in parentheses are computed based on standard errors with Newey–West correction of 12 lags.

(b) Table 8: Dual holders.

5.6 Table 9 and Table 10: analyst revisions and robustness

Read:

- Table 9 shows that RCRet positively predicts equity analyst forecast revisions in the next month.
- A 10% increase in RCRet is associated with an economically meaningful increase in next-month forecast revisions.
- Table 10 shows that the main portfolio result is robust to tighter and looser link definitions.
- The high-minus-low return remains positive under alternative stock filters, including excluding firms below the NYSE 10th or 20th percentile.

Economic message: Equity analysts also incorporate bond-link information slowly, reinforcing the market-segmentation interpretation. Robustness tests show that the results are not driven by the exact link threshold or microcap filters.

Table 9
Predicting equity analyst forecast revisions.

	(1)	(2)	(3)	(4)
DepVar:	Revision_Mean _{t+1}		Revision_Med _{t+1}	
RCRet _t	0.0074*** (3.03)	0.0052** (2.43)	0.0056** (2.39)	0.0046** (2.51)
Revision_Mean _t		0.0416*** (2.76)		
Revision_Med _t				0.0589* (1.85)
Size		-0.0000 (-0.34)		-0.0000 (-0.08)
BM		-0.0007 (-1.65)		-0.0009** (-2.12)
RVSL		0.0078*** (5.20)		0.0068*** (5.22)
MOM		0.0043** (2.15)		0.0035** (2.49)
No. Obs.	62,452	62,452	62,452	62,452
Adj. R ²	0.0068	0.1587	0.0053	0.1441

This table reports results from Fama-MacBeth forecasting regressions of future equity analyst forecast revisions. The unit of observations is at firm-by-month level, and the sample period is from Jan 2001 through Dec 2019. The dependent variable, equity analyst forecast revision for a given firm in month $t+1$, is the consensus analyst forecast of fiscal year one EPS in month $t+1$ minus consensus analyst forecast of fiscal year one EPS in month t , scaled by share price at month t end. The consensus EPS forecast is measured by the mean of EPS forecasts in columns (1)–(2); and is measured by the median of EPS forecasts in columns (3)–(4). The key independent variable is RCRet in month t . Control variables include consensus forecast revision in month t , the natural logarithm of market capitalization (*Size*), book-to-market ratio (*BM*), past one-month return (*RVSL*), and past 12-month returns skipping the most recent month (*MOM*). All variables are winsorized at 1% and 99% levels in each month. t -statistics

(a) Table 9: Analyst forecast revisions.

Table 10
Robustness: link tightness and stock filters.

Panel A: Link Tightness			
	(1) Tight Link	(2) Original Link	(3) Loose Link
1 (Low)	-0.18 (-1.21)	-0.15 (-1.10)	-0.13 (-1.70)
2	-0.09 (-0.84)	-0.20 (-1.60)	-0.05 (-0.57)
3	0.08 (0.81)	0.03 (0.25)	-0.05 (-0.56)
4	0.10 (0.72)	0.06 (0.63)	0.06 (0.62)
5 (High)	0.33 (2.47)	0.30 (2.96)	0.21 (2.45)
5-1	0.51 (2.29)	0.45 (3.47)	0.34 (3.42)
Panel B: Stock Filters			
	(1)	(2)	(3)
1 (Low)	-0.17 (-1.30)	-0.16 (-1.21)	-0.20 (-1.45)
2	-0.18 (-1.48)	-0.19 (-1.47)	-0.14 (-1.21)
3	0.03 (0.27)	-0.03 (-0.33)	-0.03 (-0.30)
4	0.08 (0.82)	0.02 (0.18)	0.00 (-0.00)
5 (High)	0.31 (2.96)	0.23 (2.03)	0.22 (2.08)
5-1	0.48 (3.69)	0.39 (3.08)	0.42 (3.41)

This table shows robustness checks for the performance of stock portfolios sorted by RCRet (Table 3) in the holding period from Jan 2001 to Dec 2019. In Panel A, we use alternative criteria for constructing the rating-comovement link. In column (1), we define a rating comovement as two firms receiving a rating co-update within a seven-day window, and we identify the firm linkage if the two firms experienced rating comovements at least three times in the past two years (Tight Link). In column

(b) Table 10: Robustness.

6 Short summary

- The paper introduces rating-comovement links as a bond-market-based measure of economic connections among firms.
- Two firms are linked if their corporate bond ratings move in the same direction within ± 10 days at least three times in the past three years.
- Linked firms comove in profitability, sales, EPS, new capital, and credit-risk measures.

- Peer-firm stock returns based on these links predict focal-firm stock returns.
- A high-minus-low stock portfolio based on RCRet earns a six-factor alpha around 0.45% per month value-weighted and 0.56% equal-weighted.
- Fama–MacBeth regressions confirm the cross-sectional return predictability after controls.
- Predictability is stronger when information flows from large peer firms to small focal firms.
- Spanning regressions show that rating-comovement momentum is not explained by other economic-link strategies.
- The effect is weak in the bond market, consistent with bond investors already pricing the information.
- Dual holders weaken stock return predictability, suggesting that investors who operate in both markets reduce segmentation.
- Equity analysts revise forecasts slowly in response to RCRet, showing delayed incorporation of bond-market link information.
- The broad lesson is that bond-market information can reveal economic links that equity markets incorporate only gradually.

7 Conclusion

7.1 Core conclusion

Feng, Huo, Liu, Mao, and Xiang show that corporate bond rating comovement reveals economically meaningful connections among firms and that these connections generate cross-stock return predictability. The central empirical fact is that peer-firm stock returns linked through repeated rating comovement forecast focal-firm stock returns.

7.2 Economic intuition

Credit-rating updates aggregate bond-market information about firm fundamentals and default risk. When two firms repeatedly receive same-direction rating changes close in time, they likely share credit-risk exposures or economic fundamentals. Bond investors and rating agencies recognize this link, but equity investors and analysts do not immediately incorporate it. This segmentation creates a delayed stock-price response.

7.3 Takeaway

The paper adds a cross-asset information channel to the economic-links literature. Economic links can be discovered not only from supply chains, industries, analysts, geography, or technology, but also from corporate bond rating behavior. Because equity and bond markets are segmented, these bond-market links produce profitable and persistent cross-stock return predictability.